

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#2} \\ \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline \mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#2} \\ \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \mathcal{K}_{1^+}^{\#1} & \mathcal{K}_{1^+}^{\#2} \\ \hline -\frac{\gamma_1 k^2}{3} & -\frac{1}{3} \sqrt{5} k^2 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline \mathcal{K}_{1^+}^{\#1} & \mathcal{K}_{1^+}^{\#2} \\ \hline -\frac{1}{3} \sqrt{5} k^2 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} & -\frac{\gamma_2 k^2}{3} \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline \mathcal{K}_{2^+}^{\#1} \\ \hline -\gamma_3 k^2 \\ \hline \end{array} \\
\mathcal{K}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline \mathcal{K}_{3^+}^{\#1} \\ \hline -2\gamma_3 k^2 \\ \hline \end{array}
\end{array}$$

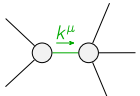
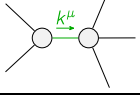
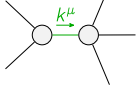
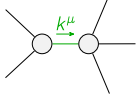
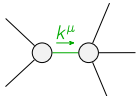
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#2} \\ \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline \mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#2} \\ \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \mathcal{J}_{1^+}^{\#1} & \mathcal{J}_{1^+}^{\#2} \\ \hline -\frac{\gamma_2 k^2}{3 \text{Det}(1^-)} & \frac{\sqrt{5} k^2 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}}{3 \text{Det}(1^-)} \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline \mathcal{J}_{1^+}^{\#1} & \mathcal{J}_{1^+}^{\#2} \\ \hline \frac{\sqrt{5} k^2 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}}{3 \text{Det}(1^-)} & -\frac{\gamma_1 k^2}{3 \text{Det}(1^-)} \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline \mathcal{J}_{2^+}^{\#1} \\ \hline -\frac{1}{k^2 \begin{smallmatrix} (4) \\ \kappa_1 + \kappa_2 \end{smallmatrix}} \\ \hline \end{array} \\
\mathcal{J}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline \mathcal{J}_{3^+}^{\#1} \\ \hline -\frac{1}{2k^2 \begin{smallmatrix} (4) \\ \kappa_1 + \kappa_2 \end{smallmatrix}} \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\gamma_1 == 2 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} \&\& \gamma_2 == 6 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} \&\& \gamma_3 == \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} \&\& \text{Det}(1^-) == \frac{1}{9} k^4 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} (7 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix})$$

Lagrangian

$$\begin{aligned}
& \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \partial_\beta \mathcal{K}_\chi^\delta \partial^\chi \mathcal{K}_\alpha^{\beta} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} \partial_\chi \mathcal{K}_\beta^\delta \partial^\chi \mathcal{K}_\alpha^{\beta} + 3 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + 3 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \\
& 2 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \partial^\chi \mathcal{K}_\alpha^{\beta} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} \partial^\chi \mathcal{K}_\alpha^{\beta} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#2} == 0$	1	$\partial_\chi \partial_\beta \partial_\alpha \mathcal{J}^{\alpha\beta\chi} == \partial_\chi \partial^\chi \partial_\beta \mathcal{J}^\alpha_\alpha{}^\beta$	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\chi \partial_\beta \partial_\alpha \mathcal{J}^{\alpha\beta\chi} == 0$	
Total # constraint(s):	2		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	4	0	$-\frac{1}{\begin{smallmatrix} (4) \\ \kappa_1 + \kappa_2 \end{smallmatrix}}$
	4	0	$\frac{1}{\begin{smallmatrix} (4) \\ \kappa_1 + \kappa_2 \end{smallmatrix}}$
	2	0	$\frac{1}{7 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 + 8 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2} \text{Root}[-508032 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^3 - 653184 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} - 155520 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2 - 10368 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^3 + (26208 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 + 17856 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} + 2016 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2) \#1 + (-328 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} - 88 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}) \#1^2 + \#1^3 \&, 1]$
	2	0	$\frac{1}{7 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 + 8 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2} \text{Root}[-508032 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^3 - 653184 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} - 155520 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2 - 10368 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^3 + (26208 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 + 17856 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} + 2016 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2) \#1 + (-328 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} - 88 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}) \#1^2 + \#1^3 \&, 2]$
	2	0	$\frac{1}{7 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 + 8 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} + \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2} \text{Root}[-508032 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^3 - 653184 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} - 155520 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2 - 10368 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^3 + (26208 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix}^2 + 17856 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix} + 2016 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}^2) \#1 + (-328 \begin{smallmatrix} (4) \\ \kappa_1 \end{smallmatrix} - 88 \begin{smallmatrix} (4) \\ \kappa_2 \end{smallmatrix}) \#1^2 + \#1^3 \&, 3]$
Resolved unitarity condition(s):	(Demonstrably impossible)		